

FULL NAME: _____

BLOCK: _____

Function Operations + Composition Order DOK-3 (Standalone)

Lesson 5-5 · Quick · Student Packet

OBJECTIVES

Math Objective	Combine functions via $+$, $-$, $\times$, $\div$, and composition; track how domain restrictions change; recognize that composition is non-commutative and explain when the order of operations changes the outcome.
Language Objective	Justify in writing which order of a two-step discount gives a better price, using function composition.
Essential Understanding	$f \circ g \neq g \circ f$ in general — the ORDER of composition matters and often changes the answer, which matters for real-world decisions.

TOPIC GOALS / ESSENTIAL QUESTION / MATERIALS

Topic Goals	Add, subtract, multiply, divide, and compose functions; recognize non-commutativity of composition.
Essential Question	Why does the ORDER of a two-step discount or two-function composition change the result, and when does it not?
Materials	Student packet, pencil. DOK-3 Practice #32 (music store) is the lesson's standalone reasoning capstone — no Topic 5 assessment overlap, treated on its own merits.

LAUNCH — Function Operations + Composition (teacher models Ex 1 & Ex 4)**[BOOK] FUNCTION OPERATIONS RULES (keep printed on your page)****FUNCTION OPERATIONS**

- $(f \pm g)(x) = f(x) \pm g(x)$; $(fg)(x) = f(x) \cdot g(x)$; $(f/g)(x) = f(x)/g(x)$, $g(x) \neq 0$.
- Domain of combined function = intersection of domains, minus any new excluded values (e.g., $g(x) = 0$ for quotient).
- Composition: $(f \circ g)(x) = f(g(x))$ — apply g FIRST, then f .
- $f \circ g \neq g \circ f$ in general. ORDER MATTERS. Always check with a test value if unsure.

[WARN] COMPOSITION IS NOT COMMUTATIVE

$(f \circ g)(x) = f(g(x)) \neq (g \circ f)(x) = g(f(x))$ in general.

Step 1: identify which function is applied FIRST (the inner function).

Step 2: substitute the entire inner function as the input to the outer function.

Step 3: simplify. Then check: would reversing the order change the answer?

EXPLORE — Think-Pair-Share (33 min)

Work with a partner. Move in order. Practice #32 (Music Store) is the big one — plan ~12 min for it. 45-min Wed F variant: skip q34 and q28.

Bank-item labels (in order):

1. Try It 1 — Add/subtract functions
2. Try It 4 — Compose functions
3. Practice #20 — Function operations
4. Practice #21 — Function operations
5. Practice #24 — Domain of combined function
6. Practice #28 — Composition evaluation
7. Practice #34 — Mixed operations
8. Practice #32 [STAR] DOK-3 ORDER-MATTERS — Music Store Double Discount

Try It 1 — Add/subtract functions

Let $f(x) = 2x^2 + 7x - 1$ and $g(x) = 3 - 2x$. Identify rules for the following functions.

- a. $f + g$
- b. $f - g$

Try It 4 — Compose functions

Let $f(x) = 2x - 1$ and $g(x) = 3x$. Find the value or identify the rule for the following functions.

- a. $f(g(2))$
- b. $f(g(x))$

Practice #20 — Function operations

Let $f(x) = \frac{5}{3}x^2 + 2x - \frac{5}{8}$ and $g(x) = 3x^2$. Identify the rules for the following functions. SEE EXAMPLE 1

$f + g$

Practice #21 — Function operations

Let $f(x) = \frac{5}{3}x^2 + 2x - \frac{5}{8}$ and $g(x) = 3x^2$. Identify the rules for the following functions. SEE EXAMPLE 1

$f - g$

Practice #24 — Domain of combined function

Let $f(x) = 4x - 5$ and $g(x) = -7x$. Evaluate each expression. SEE EXAMPLE 4

$f(g(3))$

Practice #28 — Composition evaluation

Let $f(x) = x^2 + x$ and $g(x) = 9 - 2x$. Identify the rules for the following functions. SEE EXAMPLE 5

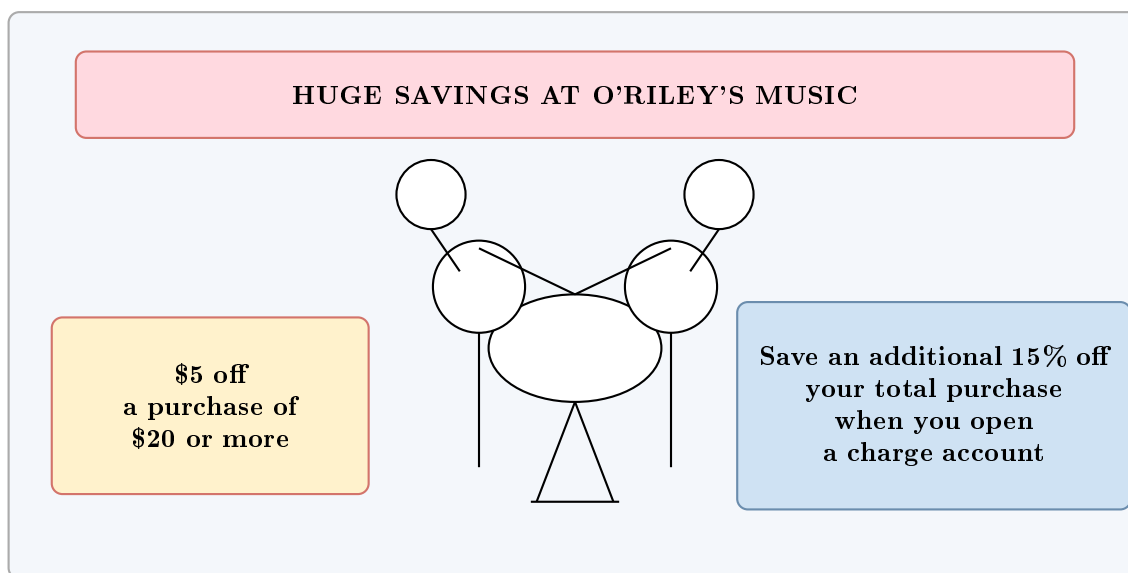
$f \circ g$

Practice #34 — Mixed operations

Given that $f(x) = x^2 + 8x + 3$ and $g(x) = -x - 7$, which of the following are true? Select all that apply.

- $f + g = x^2 + 7x - 4$
- $f(g(x)) = x^2 + 6x - 4$
- The domain of $(f)/(g)$ is the set of all real numbers.
- $f(x) \cdot g(x) = -x^3 - 15x^2 + 53x + 21$
- In the composition $g \circ f$, the output $f(x)$ is used as the input for g .

Practice #32 [STAR] DOK-3 ORDER-MATTERS — Music Store Double Discount
Use Structure A music store is running the following promotions.



- a. Use composition of functions to find the sale price of a \$90 purchase when the \$5 off discount is applied prior to the 15% off discount.
- b. Use composition of functions to find the sale price of a \$90 purchase when the 15% off discount is applied prior to the \$5 off discount.
- c. In which order is the deal better for the consumer? Explain.

SHARE / SUMMARY (5 min)

Sentence frames

Pair share (#32 Music Store): “Applying ____ first gives ____, then ____ gives _____. If I reverse the order I get ____, so $(f \circ g)(x) = ___ \neq g \circ f = ___$.”

Whole class: one pair presents their order-matters argument. Class signals thumbs up/sideways.

[SEARCH] LESSON WRAP

This is a standalone lesson — composition order is the key idea to carry forward. It appears in real decisions: discounts, conversions, multi-step processes.

EXIT

$f \circ g$ means ___ is applied ___.

An example where $f \circ g \neq g \circ f$ is ___.

For a double discount, the better order for the consumer is ___ because ___.

One biggest thing I learned today: _____

OPTIONAL REINFORCEMENT (not collected)

If you finish Explore early. #35 is a SAT/ACT-style item requiring you to evaluate $f(g(5))$.

Optional #1 (Savvas Practice, SAT/ACT $f(g(5))$)

SAT/ACT Find the value of $f(g(5))$ if $f(x) = 4x + 1$ and $g(x) = x^2 + 6$.

A 101 **B** 124 **C** 125 **D** 676 **E** 682